"AEOLUS"
A System Concept for Precise Delivery of Mars Scientific Instrumentation

prepared by

Marcus S. Murbach*
NASA Ames Research Center, MS 244-14
Moffett Field, California
(415) 604-3155
mmurbach@mail.arc.nasa.gov

David L. Keese†
Sandia National Laboratories
P.O. Box 1500, MS - 0313
Albuquerque, New Mexico 87185
(505) 844-1899
dlkeese@sandia.gov

JackFarmer‡
NASA Ames Research Center
Moffett Field, California, MS 239-4
(415) 604-5748
jfarmer@mail.arc.nasa.gov

Abstract. The AEOLUS system concept is proposed to deliver science instruments with great precision to the surface of Mars. This capability is regarded as an important next step in bringing Mars evolutionary processes into greater focus by allowing investigation of specifically targeted sites. By the use of existing maneuvering vehicle technology, terrain-guidance hardware, and other vehicle subsystems this mission appears feasible with a minimum amount of development and risk. The conceptual system design presented in this paper is capable of delivering a variety of instrumentation types (the first phase is a penetrator/flight experiment, followed by a rover mission) with 3 sigma accuracies of approximately 1 km compared with the typical Mars Pathfinder landing ellipse of 150 km. This allows the utility of small rovers to greatly increase as the goal is to develop a system such that the rover range exceeds the landing ellipse of the delivery system. The proposed mission using available technologies in combination with a novel approach to transportation, results in attractive mission costs from $35M-$50M. The system name "AEOLUS" (Greek god of the air and wind) was chosen since it would represent controlled flight in another planetary atmosphere.

Introduction

Mars is a particularly fascinating area of investigation due to the long early history during which benign, earth-like conditions prevailed for perhaps many hundreds of millions of years. The record of that evolution, as well as the possibility that life once existed, is locked within the mineralogical/hydrological history of the planet (Reference 1). The suggestion that complex organic processes were preserved in the Mars-originated ALH84001 meteorite is still being debated by the community (Reference2). More recently, the success of the Mars Pathfinder (the Sagan lander) and

* Research Scientist, Space Projects Division, Member AIAA
† Manager, Rocket Systems Technology Department
‡ Research Scientist, Space Projects Division, Exobiology Branch

associated Sojourner rover has helped stimulate further investigations. A dominant theme in the AEOLUS mission proposal is "bringing Mars into greater focus" by incorporating a unique combination of available technologies.

Mission Objectives

The AEOLUS mission is designed to demonstrate the precision delivery of small science payloads to the surface of Mars with an accuracy that is unrivaled by current systems. It builds on current precision delivery systems developed for the Department of Energy that can achieve, on earth, an accuracy of ~3 meters (depending on the resolution of available maps). Precision targeting of landers to high priority geologic sites on Mars has been identified as a fundamental technology issue that must be addressed if we are to accomplish the present science goals of the Integrated Mars Exploration Program, namely the search for past or present life, understanding the climate and volatile history of Mars, and assessing the resource base for future robotic and human exploration. Because these science objectives are highly site specific, future surface missions must:

- increase the capability of rovers to reach targets of interest during nominal missions, and
- increase landing precision to place landers and rovers within reach of specific targets of interest as defined by rover mobility.

Presently the targeting errors for landed missions are on the order of 150 km, which greatly exceeds the reach of the present generation of microrovers during nominal surface mission times. Thus present delivery systems are only appropriate for addressing science questions that are not site specific or involve targets for study that are large and homogenous. The AEOLUS delivery system will greatly improve targeting accuracy for small landed science packages and is capable of placing payloads within 50-500 m of pre-specified geologic targets. This accuracy is well within reach of current microrover platforms which are capable of 1 km traverses within mission times of a few months (the mobility for the 2001 rover technology is intended to be several km).

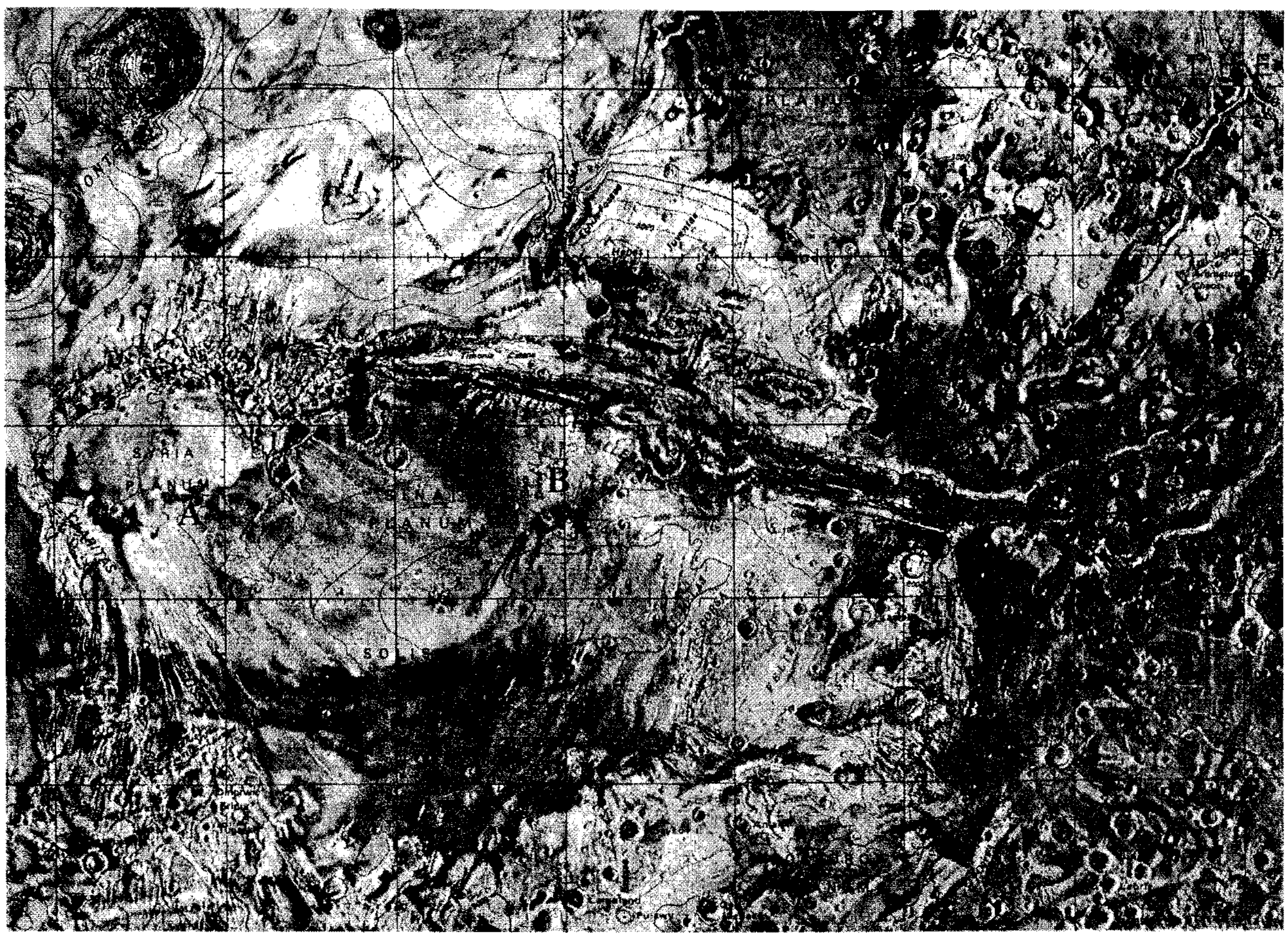

Figure 1. Valles Marineris

The proposed mission involves an initial flight technology demonstration phase that will demonstrate the feasibility of the low-cost, high precision AEOLUS delivery system. In this initial phase, only comparatively simple science goals will be addressed, emphasizing the use of existing technologies to demonstrate precision delivery to pre-specified targets on the ground. This will be accomplished by dropping clusters of penetrators within close proximity of several pre-determined high priority targets along the mission flight path.

Science Rationale

The proposed site to be explored during the mission is the Valles Marineris, the largest canyon in the solar system, is shown in Figure 1. This canyon, represents one of the most significant Martian geological features and can only be explored with precision navigation. This extraordinary feature, is approximately 3500 km long, 20-200 km wide and 6-9 km deep, and has evoked intense interest since it was first photographed from orbit by Mariner 9. The giant equatorial canyon and its associated chasmata lie along an extensive radial fault system which extends eastward from the Tharsis volcanic province, the locus of extensive late Hesperian to Amazonian volcanic activity and site of the largest volcanoes in the solar system (Reference 3).

The overarching science goal of the proposed mission is to evaluate theories for the origin of Valles Marineris, its aqueous history, and potential as an abode for ancient Martian life. The relative roles of mass wasting and transport by water as geologic agents in the formation of the canyon system remain controversial. We propose to investigate the processes of Canyon formation by:

- mapping the mineralogy of canyon and floor deposits along the mission flight path using UV remote sensing methods (using a DASI, or Digital Array Scanning Interferometer, described in Reference 4) in order to determine the nature of past aqueous processes,
- measuring the surface pressure and the volatile composition of deposits over a range of elevations from the canyon rim to floor to evaluate the role of subsurface ices (including CO_2 clathrates) in mass wasting of the canyon walls, and
- using penetrators equipped with sensors to measure temperatures and thermal conductivities of the Martian regolith; to determine the thermal properties of canyon floor and wall deposits to examine the physical properties of the regolith as a basis for better understanding the retention of volatiles; and to investigate the depth and distribution of ground ice with elevation, and which will establish the potential for a subsurface hydrosphere.

Penetrators designed for different depths (up to 24 penetrators at 2 and 4 m depths) will be delivered to at least three pre-determined locations along the flight path (Figure 2). These devices will be deployed at sites along the canyon rim, slope and floor, and at two different depths at each site. Volatile information will be acquired using soil samples captured by the penetrators upon impact. These samples will subsequently be heated to several hundred degrees Celsius and the evolved gases identified using a tunable laser diode array (Reference 5). A pressure sensor on each penetrator will be active throughout the mission and providing measurements of atmospheric pressure with an accuracy of 0.03 mbar. The above information will provide an empirical framework of physical/chemical constraints for evaluating previously proposed hypotheses regarding the relative roles of aqueous vs. mass wasting processes in the formation of Valles Marineris; for testing theoretical models that predict the latitudinal changes in cryosphere depth (in relationship to regolith permeability); as well as the existence of a subsurface groundwater (and possibly hydrothermal) system in association with the present canyon system.

In addition to addressing the fundamental science goals outlined above, the penetrator instrumentation package will also provide a basis for evaluating technological performance of penetrators and the engineering properties of the Martian regolith as a basis for future refinements in penetrator design. In that regard, descent and impact accelerometers will provide important data concerning atmospheric drag forces acting on the aeroshell during descent and deceleration forces acting on the penetrator forebody during impact.

Aeolus Mission Architecture

Penetrator Mission

The technology/penetrator mission is summarized in Figure 2 (further details of the scenario are provided in subsequent sections). The AEOLUS stack will be placed into LEO (Low Earth Orbit) as an STS (Space Transportation System) piggy-back mission several months before the Mars injection opportunity (the April 2000 Type II Mars opportunity was used). At that time, the system will first be properly positioned with the 3-axis stabilization system. It will then be spin-stabilized and a Star 30 will perform the Trans-Mars Injection. Several months later, the cruise stage will provide mid- and final course corrections, whereupon the 178 kg AEOLUS system will be separated from the cruise system. After initial atmospheric entry the vehicle will perform a pull-up maneuver and controlled flight will commence. The vehicle will be at point "A" in Figure 1 (Syria Planum) and fly eastwards over the canyon at point "B" and the outflow channel, point "C". The on-board terrain navigation system will maintain the vehicle at a constant altitude of 5 km above the rim of Valles Marineris during the 20 minutes it will take to traverse the length of the canyon. During the flight phase, penetrators are ejected at regular intervals at predetermined locations.

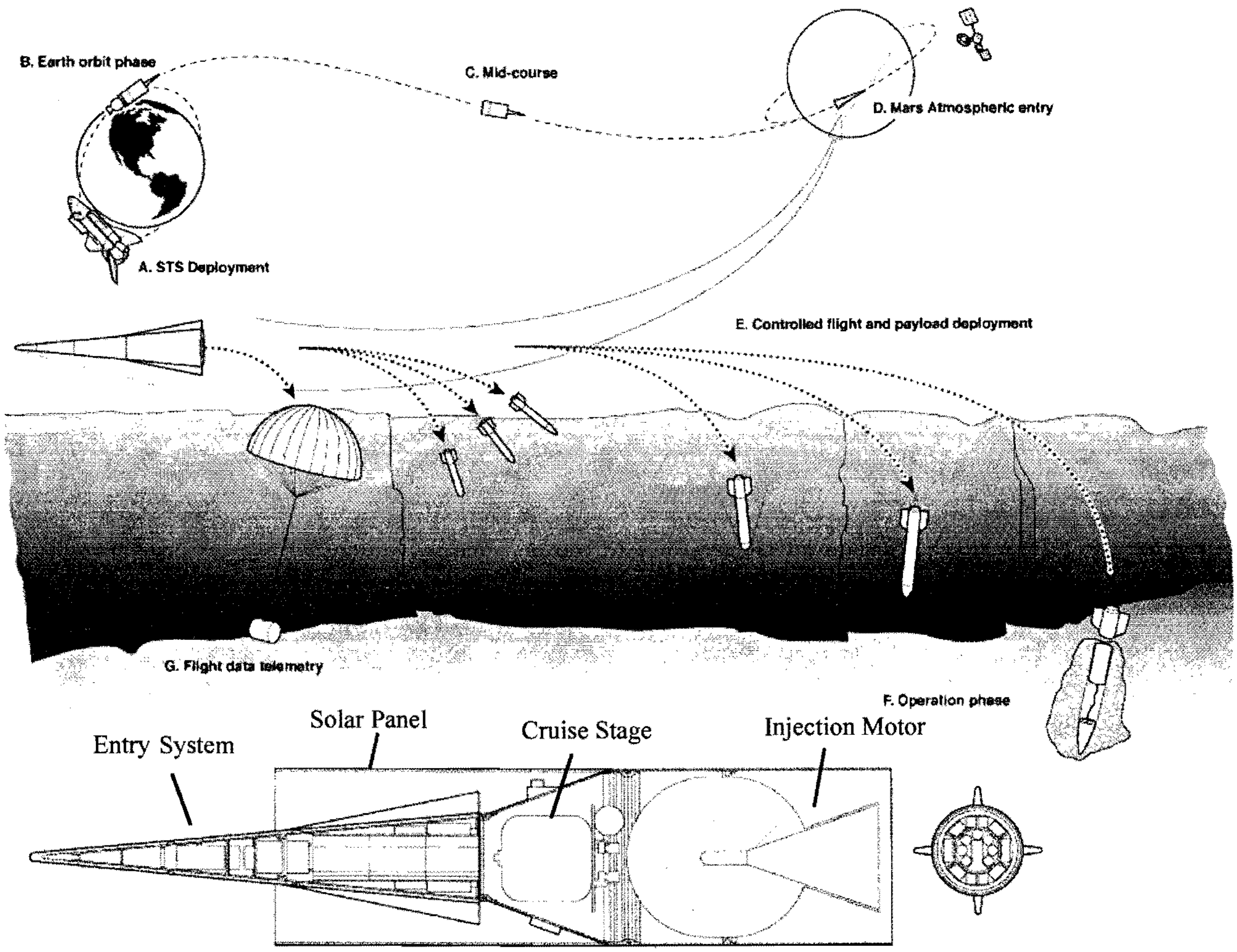

Figure 2. AEOLUS Technology Mission and Vehicle Stack Description

Rover/Imaging Mission

The AEOLUS rover mission is similar to the penetrator mission except that the penetrator carousels are replaced by microrover canisters, as shown in Figure 3 (References6,7). At two pre-determined locations a canister containing a communication package and instrumented micro-rover will be deployed to the canyon bottom. Also, 5 m resolution UV images will be obtained with a DASI (Digital Array Scanning Interferometer)and the data stored in the forward most canister. At the end of the flight this final canister containing the flight data will be parachuted and the vehicle discarded (crash landed). The three landed science packages will then begin telemetry to a Mars Global Surveyor orbiter relay. Constraints for this flight are that the overflight velocity is slower, and that it occurs in the daylight for imaging purposes (unlike the initial mission).

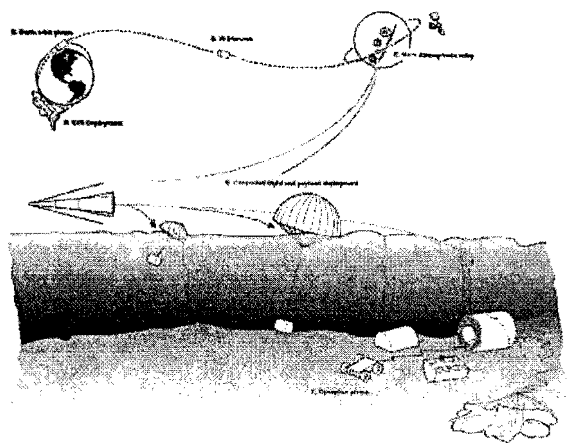

Figure 3. AEOLUS Mission with Evolved Rover Option

Flight System Design Concept

Flight System

An enlarged view of the complete AEOLUS flight system is shown in Figure 4. The stack is intended to fit vertically in the shuttle cargo bay for deployment in low earth orbit (LEO). The overall vehicle stack is 1.2m in diameter by 4.5m in length with a total mass of 778 kg. The Star 30 is the Trans-Mars Injection stage located at the aft end. The cruise stage, comprised of a modified Precision Injection kit, provides the 3-axis stabilization in LEO (prior to spin stabilizing the Star motor). The cruise stage also performs the mid and final course corrections during the Mars trajectory. On the side of the thrust structure housing is mounted the Star-tracker navigation subsystem as well as the power and communication systems. On the exterior is a cylindrical solar panel which provides sufficient power system regeneration at low angles of incidence. It was desirable to eliminate deployable panels due to spin up/down issues. This structural element is visible in Figure 2.

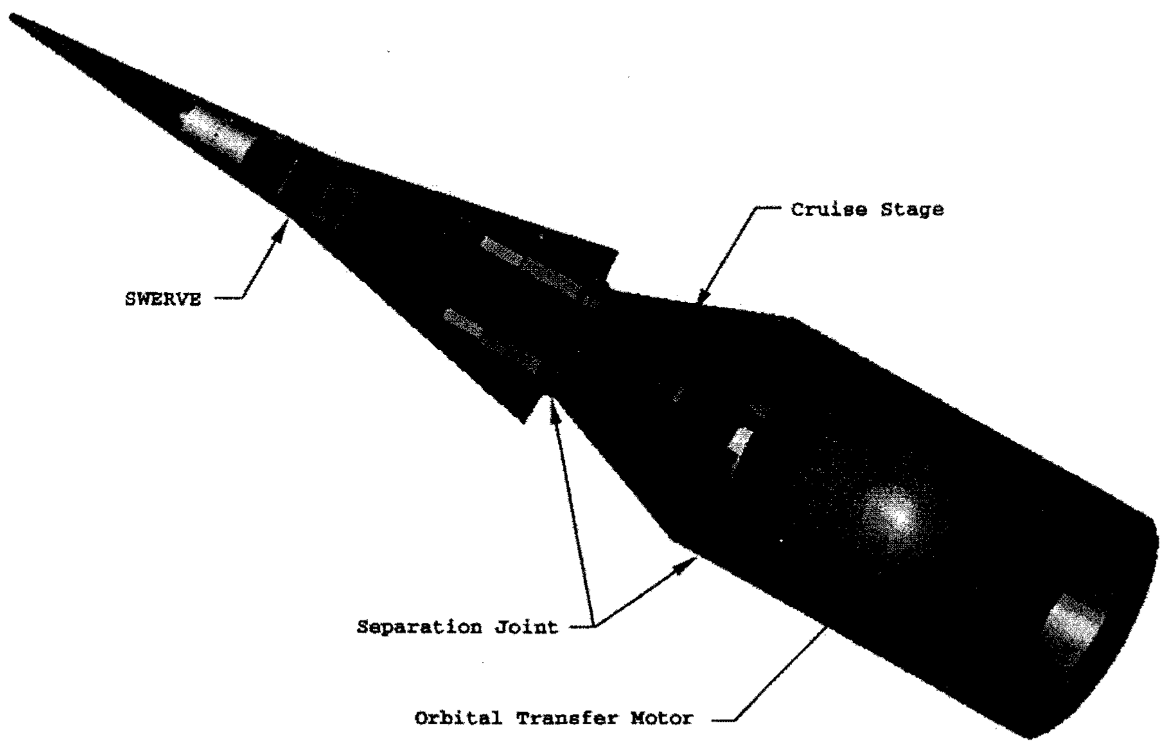

Figure 4. AEOLUS Flight System Stack (solar panel structure not shown)

Entry System

A key feature of the AEOLUS system concept is the use of existing maneuvering vehicle technology developed for the Department of Energy by Sandia National Laboratories. The SWERVE-derived (Sandia Winged Energetic Reentry Vehicle) system shown in Figure 5 was originally developed and flight tested as a highly maneuverable weapon system. Modifications for use in the Mars entry environment have been studied and will be presented later in this paper. The 2.75m baseline design has an entry mass of 178 kg with a 1.2m by .25m payload section (comprising the 3 carousel/canisters). The basic design is a sharp 5° half-cone with four wings arranged in a cruciform resulting in a high L/D ratio (L/D>1). Control surfaces (controlled by electro-mechanical actuators) located on the wings provide the great maneuverability of the system. The space-qualified SANDAC V flight computer receives inputs from the navigation system, including the Inertial Terrain Aided Guidance (ITAG) system mounted in the vehicle. Power for the system is provided by flight batteries located in forward and rear compartments.

The entry system design is based on existing technology developed at Sandia as a result of over 100 reentry objects in approximately 60 missions during the last 30 years. These include complex maneuvering reentry vehicles, light-weight yet highly instrumented bodies and several recovered reentry systems. Sandia is a multi-disciplinary engineering organization operated for the Department of Energy in the mode of a Federally Funded Research and Development Center (FFRDC).

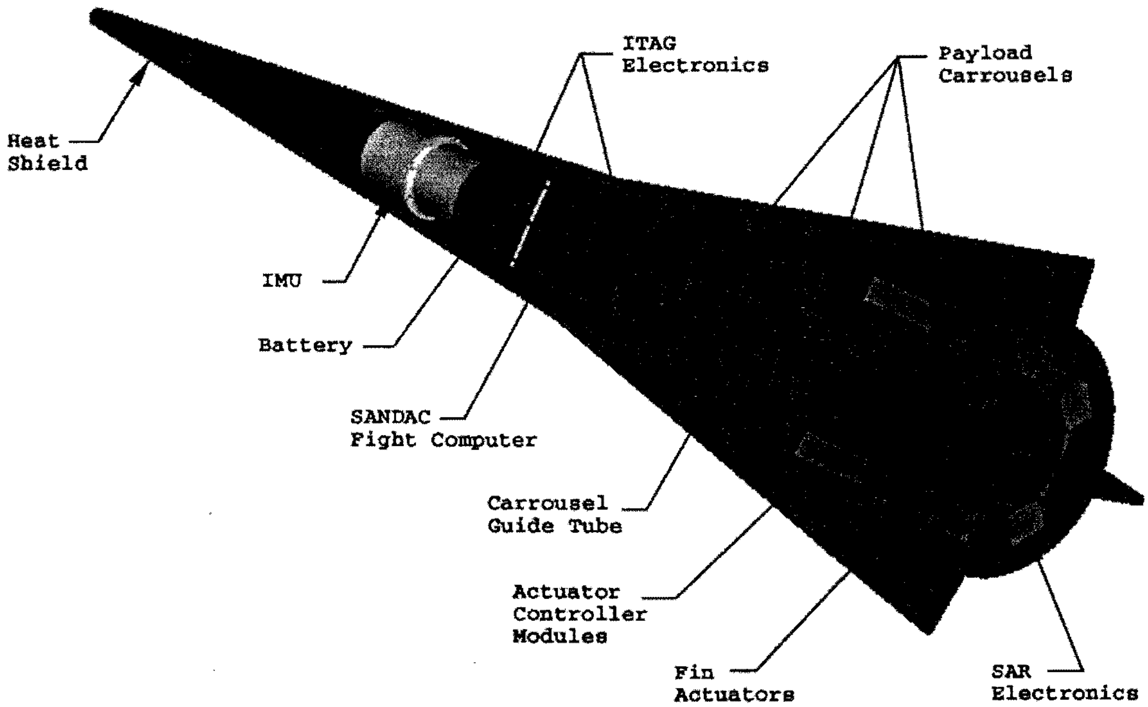

Figure 5. SWERVE Derivative Mars Entry System

Navigation, Guidance & Control

A unique feature of the proposed system is the combination of high maneuverability and the existence of flight qualified, radiation-hardened navigation system referredto as ITAG, or Inertial Terrain Aided Guidance (Reference 8). The system uses a radar altimeter for terrain elevation profile measurement and a stored Digital Terrain Elevation Data base (DTED) for terrain elevation profile prediction (the Mars terrain maps available from Mars Global Surveyor would be converted into the desired format). Once the vehicle begins controlled flight, signals from the on-board radar are compared to the stored map data base, and the vehicle navigation is updated.

Instrument release is also controlled using navigational data to accurately command payload release. Originally designed for a CEP (Circular Error Probable) of 3 m, the delivery of the package (formerly a munition) is a function of the stored map accuracy, the vehicle velocity along a particular trajectory, and how quickly the vehicle establishes a location fix after the pull-up maneuver. Current estimates are that 1 km landing ellipse is straightforward with 50 m accuracy achievable. The availability of several areas of high relief in Mars target area greatly enhances the probability of a rapid, accurate navigation fix.

The ITAG system and its general function are illustrated in Figure 6. A terrain map of the type used by ITAG is shown in Figure 7. Sufficient volume and mass have been reserved as part of the entry system preliminary design to accommodate ITAG components.

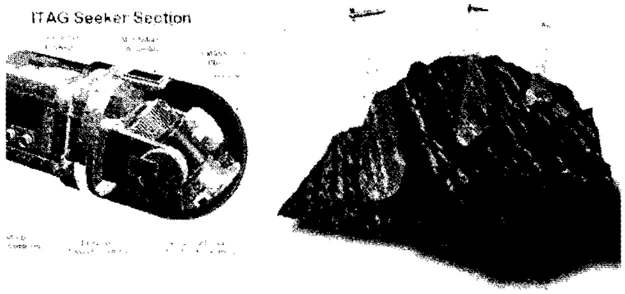

Figure 6. ITAG System Description

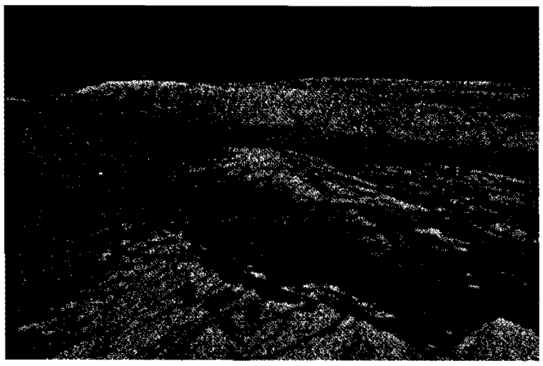

Figure 6. Terrain Map

Payload Suite (Penetrator Element)
The primary instrumentation payload proposed for the initial AEOLUS mission consists of a series of miniature penetrators. During the controlled flight phase, 24 penetrators are released from the three payload canisters. Clusters of penetrators are released at the same time and penetrate different depths. The penetrator schematic shown in Figure 8 is similar to the initial configuration jointly developed by Sandia and JPL (Jet Propulsion Laboratory) for the Deep Space - 2 mission. These penetrators perform sample collection, volatile analysis, temperature gradient measurement, and atmospheric pressure measurement. The key difference with the current DS-2 mission is that, while the internal electronics remain much the same, the penetrator unit is deployed at a lower velocity and does not experience the high heating rates requiring extensive thermal protection.

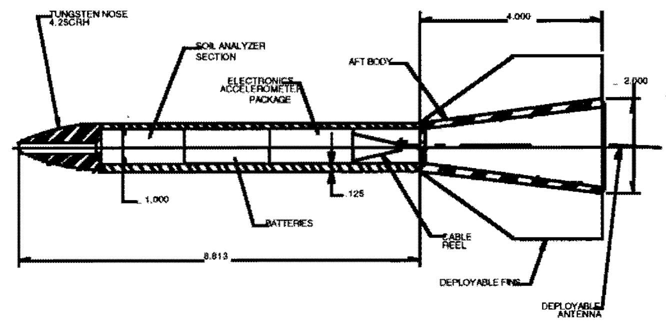

Figure 8. Penetrator Schematic

The operation of the penetrator during soil entry is illustrated in Figure 9. The detachable aft-body allows the penetrating vehicle, containing all electronics except for the antenna, to penetrate to a depth as determined by the geology. When the aft-body reaches the soil it detaches and stops at the soil surface. An umbilical line of sufficient length to cover a range of soil resistances connects the detached portion and its electronics to the aft-body and antenna. All data collected from the on-board instruments will be communicated to the surface via a spool wire deployed during the penetration event. A transmitter and antenna system will be left as the aft-body on or near the Martian surface.

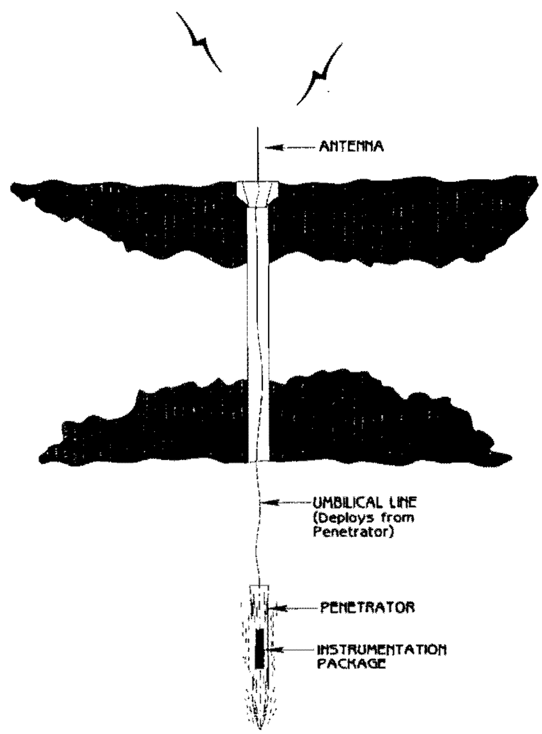

Figure 9. Penetrator Operation

The capability to successfully complete the penetrator design has been acquired through 35 years of Sandia weapon and sensor programs. This includes the ability to design and develop power systems, telemetry transmitters, accelerometers and other electronic systems required to survive this deceleration event. An indication of this extensive experience base is shown in Figure 10 (Reference 9). The vertical axis represents a target's resistance to penetration. Rock and concrete are near the bottom and very soft soils are at the top. Penetrator weights have ranged from a few pounds to a few thousand pounds. Soil types expected in this area of Mars range from an S# of 3-17 and impact velocity is expected to be ~ 200 m/sec.

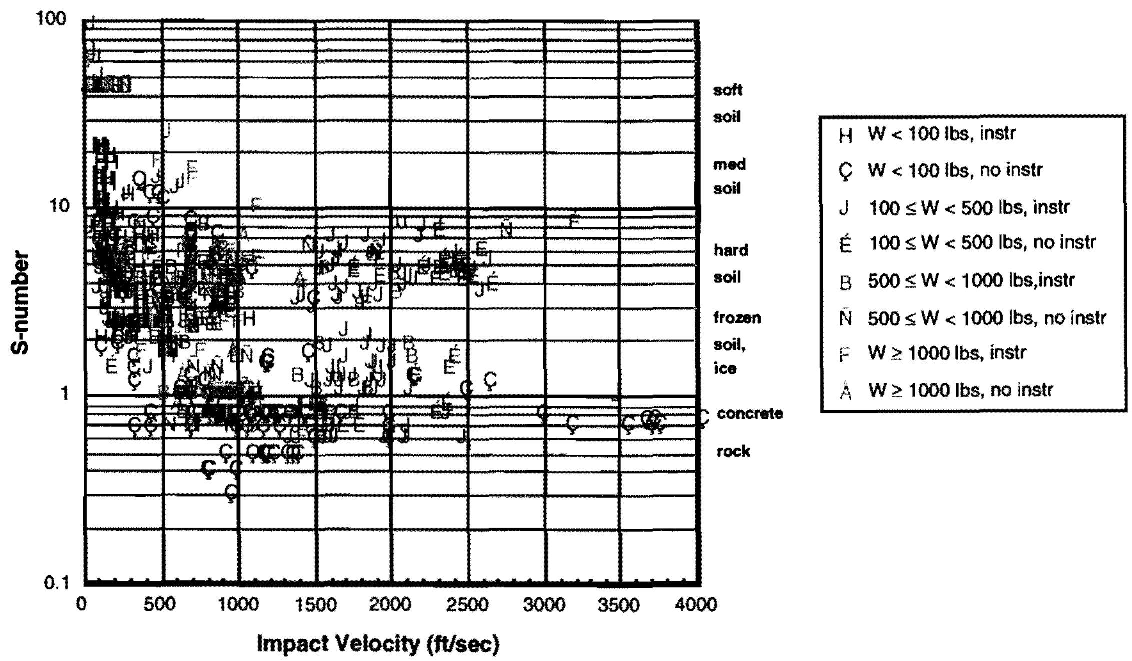

Figure 10. Sandia's Penetrator Experience

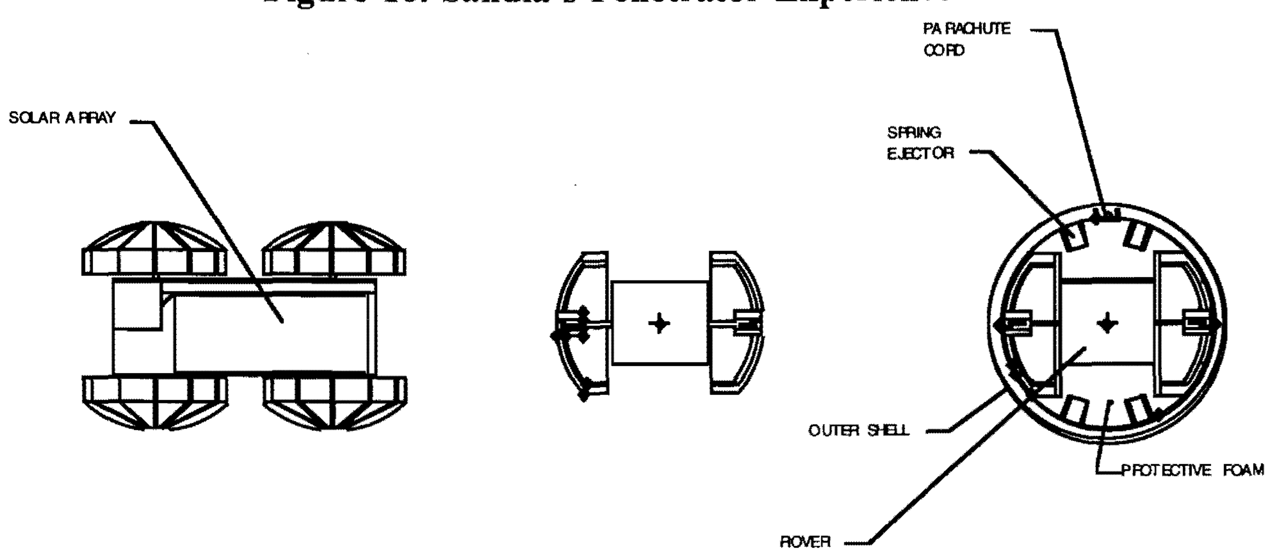

Figure 10. Rover Description and Canister Integration

Payload Suite (Rover/Imager Element)
The evolved science platform involves a 5 kg-class microrover to provide mobility for the selected science instruments. The rover and canister are shown in Figure 11, with rover dimensions of 20 x 40 cm. The canister is deployed at low supersonic speed, where it is decelerated and a parachute is deployed for final landing. The canister then strikes the surface in a controlled orientation at 20-25 m/s, bounces, and then assumes an 'arbitrary' position. At this point, the rover and packaging material are forcibly ejected and the rover 'rights' itself. A characteristic of the rover, necessary for this deployment approach, is that the rover is 'orientation independent' (i.e., the rover is designed to operate 'up' or 'down' as is found with certain radio-controlled cars which are currently popular). The power system is comprised of a primary battery system charged with both an RHU (Radioactive Heating Unit) power system and solar cells on the upper and lower surfaces. Initial deployment tests conducted earlier in 1997 (comparable impact velocity into a cement surface) proved satisfactory with the successful exit of a functional rover. As a final note, increasing the dimensions and capability of a rover are being investigated by using two standard canisters and also by modestly increasing the size of the entry vehicle.

Preliminary Analyses

Entry and Flight Profiles
In order to validate the AEOLUS concept a series of preliminary analyses were completed (References 10,11). A summary of these results are presented in this section to demonstrate the uniquely attractive features of this flight system concept and its particular application to a Mars mission. These analyses focused on general performance (range, altitude, velocity) and the aero-thermal behavior of the entry system. The entry simulations were performed with the Trajectory Simulation and Analysis Program (Reference 12) code at Sandia using the Mars GRAM (Global Reference Atmosphere Model, Reference 13). In addition, a flight simulator was modified for Mars conditions using a SWERVE-derived aerodynamic data-base. The initial conditions are described below.

Velocity = 7 km/sec
Altitude = 125 km
Entry angle = -15 degrees (below the horizon)

Mars equatorial and polar radii, rotation rate, gravity constant, and the second, third and fourth gravity potential harmonics were also used to increase the accuracy of the simulations. These simulations used an aerodynamic model of the SWERVE-derived vehicle developed using over 6000 hours of wind tunnel data that spanned the subsonic and hypersonic (Mach 22) regimes. A spot check of the aero model using a parabolized Navier Stokes code was completed to validate its applicability at the lower Reynolds Number created by the Mars atmosphere.

The initial phase of the entry simulations involved a pull-up maneuver initiated at 20 km to transition into a controlled "cruise" phase. A variety of pull-out angles of attack were examined and the results are summarized in Figure 12. The 10 degree angle of attack profile was selected for use in the subsequent phases of the study since it provided greater range capability and target selection flexibility. Two basic variations of the following cruise phase were initially examined:

- A maximum range extension and
- A profile that involved a 90 degree yaw maneuver to investigate the potential of a polar flight profile.

Figure 13 illustrates the extremely long range capability of the AEOLUS platform. This simulation indicates that the vehicle can travel for over 4000 seconds and 12,000 km from the entry point before loosing energy sufficient for controlled flight. The "right turn profile" summarized in Figures 14 and 15 show convincing evidence that the AEOLUS system can reach almost any point on the Martian surface. In addition, results with the simulator indicated that shorter range flights were possible, though at the expense of increased heating rates.

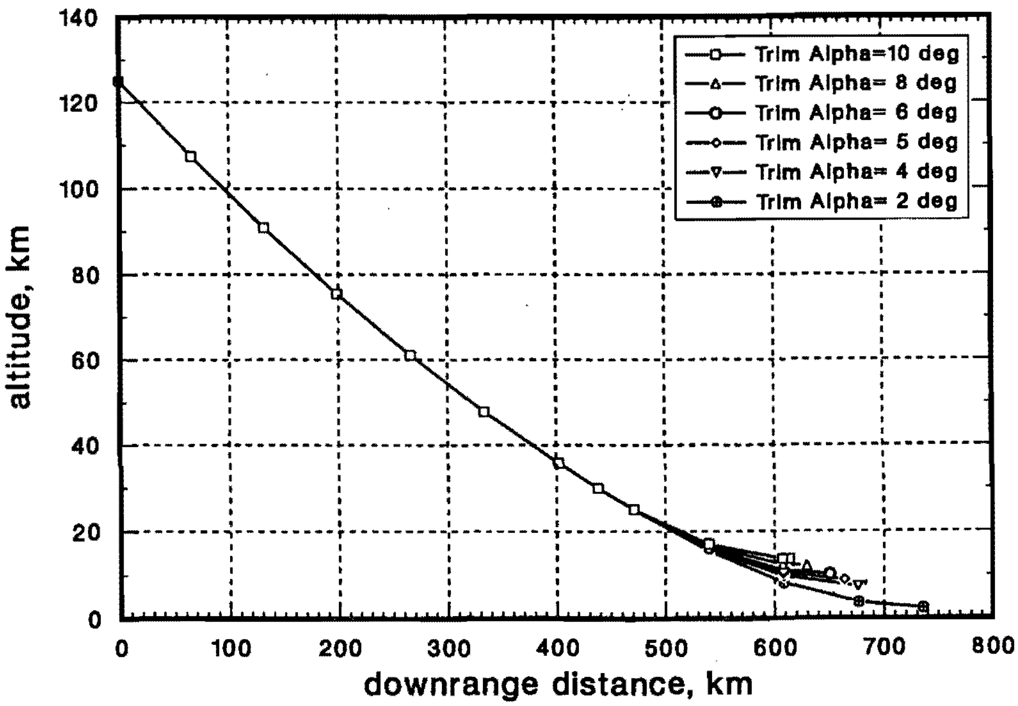

Figure 12. Flight Profile Prior to Pull-up

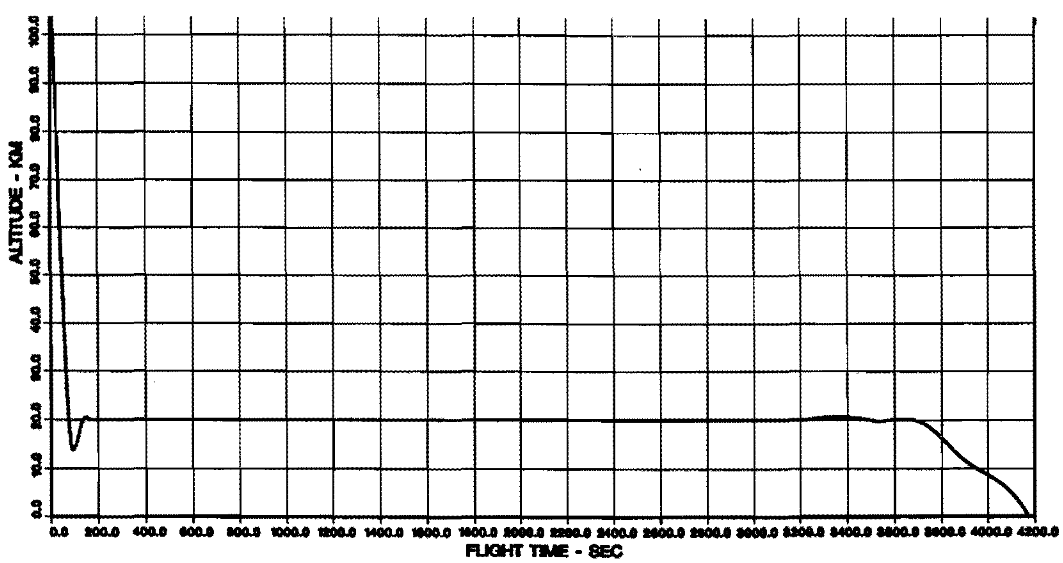

Figure 13. 20 km cruise phase

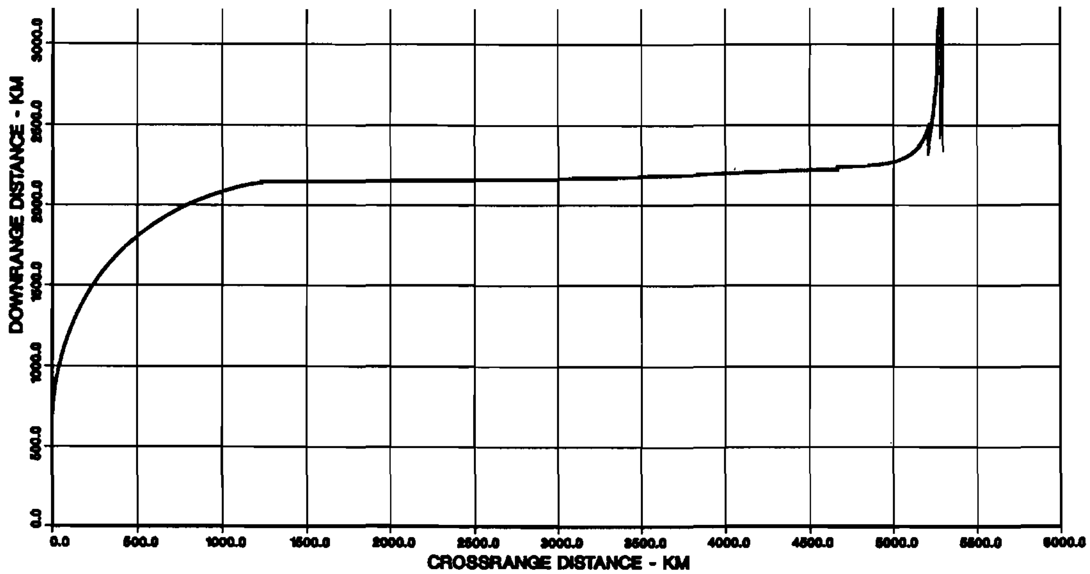

Figure 14. 90 deg Turn Profile

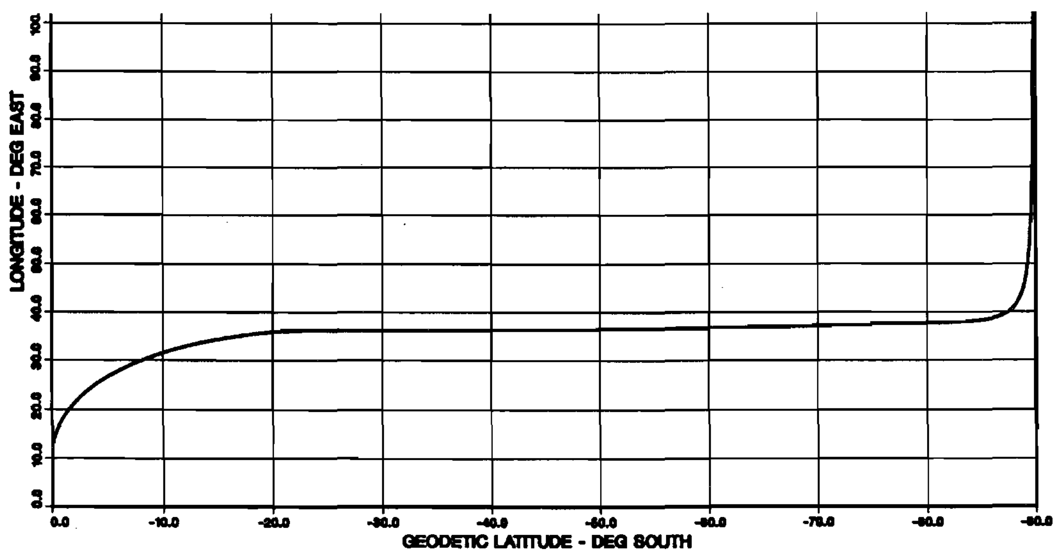

Figure 15. Latitude and Longitude

Aero-Thermal Analyses

Heating analyses were also completed along with the basic entry simulations. This stage of the feasibility assessment relied on calculated stagnation heating using analytical methods (References 14,15) and the HANDI code (Reference 16). Sidewall heating estimated as a percentage of stagnation heating and material response was examined using the Charring Material Ablation (CMA) code (Reference 17). The results of these analyses for SWERVE-derived heatshield are summarized in Table 1.

Several key aero-thermal conclusions are summarized below:

- The baseline thermal protection system (TPS) for a Mars entry should be the recently developed SIRCA (Silicone Impregnated Reusable Ceramic Ablator). This material has the advantage of tolerating the body heating rates, but the density is approximately 1/10 and the thermal conductivity is 1/9 less than the silica phenolic that is typical flight baseline (References18,19).
- A SIRCA TPS will help diminish the long heat soak time which would create adverse internal temperatures and also reduce the vehicle total mass.

- An ultra-high temperature ceramic (UHTC) developed by NASA Ames Research Center is also a candidate material for the nosetip and wing leading edges. This material has been demonstrated in both ground and flight tests to withstand temperatures of up to 3033K (5460 degrees R) without ablation (References 20, 21).

The general conclusions of these initial analyses indicate that the SWERVE-derived vehicle demonstrated in three previous flight tests can be utilized with the modification mentioned above to complete a Mars entry mission. In addition, recent work has suggested that the use of a decelerator during the initial entry phase can have a significant benefit in further reducing the thermal protection system (TPS) mass.

Table 1. Preliminary Heating Results

						Total Ablation cm (inches)		
Location	**Cross Section cm (in)**	**Max Heating Rate W/cm^2 (BTU/ft^2-s)**	**Max Pressure atm**	**Max Heated Surface Temp deg K (deg R)**	**Max Aluminum Temp deg K (deg R)**	**Pyrolysis Depth**	**Char Depth**	**Surface Recession**
Stagnation Point	22.5 (10.0) carbon - carbon	1360 (1200)	2.33	3055 (5500)	-----	-----	-----	3.43 (1.35)
Cone Sidewall	5 (2.0) silica phenolic 0.1 alum	68* (60)	2.33 *	1411 (2540)	483 (870)	2.64 (1.04)	.30 (0.12)	0.00
Wing Leading Edge	10 (4.0) silica phenolic	465 (410)	0.88	2200 (3960)	-----	2.96 (2.35)	2.79 (1.10)	1.0 (0.40)
*Cone sidewall heating rates were assumed to be 5% of the stagnation point heating. Stagnation point pressures were used in computation of cone sidewall thermal responses. Original vehicle TPS used for preliminary analysis.								

Table 2. Mass Summary

SYSTEM	ELEMENT	MASS (KG)	MASS (KG) SUBTOTAL	HERITAGE (Costing)	COMPLEXITY (Costing)
ENTRY SYSTEM					
	ITAG (Mini-rims;SAR)	4.75		Off the shelf	
	Sandac V	2.75		Off the shelf	
	Actuator	17.50		Off the shelf	
	Structure	17.00		Off the shelf	
	TPS Body (5cm SIRCA)	25.50		Off the shelf	
	TPS- Cone	2.00		Off the shelf	
	Flight Batteries	25.00	64.50	Off the shelf	
PAYLOAD SYSTEM					
	Science Payload I				
	Penetrator (DS2 mod))	1.50 (x24)		Min Mod	Low
	Canister System	4.80 (x3)		New Design	Low
			50.40		
	Science Payload II				
	Canister I (DASI/COM)	10.00		New Design	High
	Canister 2/3 (Desc Sys)	12.00 (x2)		New Design	High
	Laser Raman System	2.00 (x2)		Mod Design	Med
	Rover (str;pwr;com)	5.00 (x2)		Mod Design	Med
			48.00		
LEO/XFER/CRUISE SYSTEM					
	(Cruise/3-x Propulsion)				
	Structure	17.00		Mod Design	Medium
	Solar Panel (+str)	15.00	32.00	New Design	Medium
	(Control)			New Design	Medium
	Star Tracker	2.50			
	Sun Sensor	0.75		Off the shelf	
	Electronics	0.23		Off the shelf	
	Antenna	4.00		Mod Design	Medium
	(Trans-Mars)	1.00	24.68	Minor Mod	Medium
	Star30 (+kit)	543.00		Off-the shelf	
			599.68		
	Entry System Mass I		114.90		
	Total Mass I		714.58		
	Entry System Mass II		112.50		
	Total Mass II		712.18		

Cost and Mass Summary

The mass summary is seen in Table 2. Particularly for the entry and payload system contribution (Payload I is the penetrator, and Payload II is the rover mission), it can be seen that all of the critical components are derived from a large experience base. The mass estimates for the rover component come from both Ames and JPL experience. Cost estimates for the modified SWERVE, including upgraded TPS is $21M. The payload system cost was estimated at $6M, with the differenceof $15M for the more complex rover mission. Estimates for the LEO-loiter/Cruise system was $8M, though this depended on the complexity of the cruise stage (there are other liquid propulsion assets which are higher performance, however they are more difficult to manifest in the transportation system). In order to achieve the total mission costs mentioned previously, the transportation costs to LEO were assumed to be those of a piggy-back (see below) and treated as a Spartan-class program payload. If STS transport were included and fully costed, then the total mission cost would increase by approximately $25M.

Launch and Delivery Options

One of the attributes of the current "Discovery" series is the relatively high total transportation costs. For several proposals approximately $50M was required for the typical Delta launch vehicle and another $50M was required for the Mars injection/cruise stage designed by a large aerospace company. Nearly $100M in total transportation costs were required for missions ranging from $160-200M. This implies that high transportation costs (and the relatively high investment that it represents) reduce the total number of missions that are selected and encourages a great deal of conservatism in the choice of technology.

While an expendable launch vehicle (in this case, the relatively smaller Orbital Sciences Corporation Taurus vehicle) is also being considered, the STS Shuttle option appears most attractive. In the next few years, a quarter of the Shuttle fleet (i.e., Columbia) can't directly support Space Station operations due to the higher weight of the vehicle and greater energy requirement to achieve the 51° orbit. It will thus be relegated to science related missions. Due to the size and mass of the AEOLUS stack configuration (the mass is 778 kg and dimensionally can fit within 1.3 m of the cargo bay) and the choice of STS-rated components, the stack is "manifestable." The other critical point is to divorce the sometimes changeable STS launch schedule from the planetary launch window by placing the AEOLUS stack in low earth orbit (LEO) many months in advance in a 'LEO loitering' phase. The cost of such a piggy-back opportunity could be minimal. Even if the $5.54k/kg ($12k/lbm) advertised STS launch cost is considered, the resulting $24M cost is still less than the Taurus launch vehicle. Conversely, there is an 'opportunity' cost if this is not considered. If this is combined with a simple, STS compatible cruise stage then planetary mission costs could be significantly reduced to a level approaching $50M. This class of mission would be intended to be complementary (though allowing higher risk) to the existing Discovery series in much the same way that the Small Explorer (SMEX) is complementary to the Medium/Large Explorer program. Many more small planetary missions might be considered if this were further developed, perhaps increasing the number of exploration opportunities and allowing a healthy injection of higher risk technology.

Contribution to Future Robotic and Mars Programs

Future Mars missions foreseen in the first decade after 2000 include Mars Sample Return, and missions to lay the groundwork for manned Mars missions. Two critical technology areas in support of these missions are a) precision navigation, and b) development of lifting entry (high L/D) technology. Both of these technologies in concert are required for the surface rendezvous which is an essential feature of both of the above mission classes. In addition, landing site certification for a human mission would require a 'precision' precursor mission to survey the local terrain.

In order to reduce navigation errors, individual contributions during each mission phase have been analyzed (Rerence 22). It was found that even with an L/D of 0.8 (not achievable with the current generation blunted cones), navigation errors caused by atmospheric variability would result in landing errors of 10-20 km. To reduce this further, autonomous real-time terrain matching was found to be required (which is what ITAG accomplishes). The advantage of the ITAG system is that, unlike the optical systems proposed, it is strictly radar based (a necessary feature of a munition based application). This permits obscured or night landings when appropriate to the mission.

The second technology element, the high L/D vehicle, represents a significant departure from the current low L/D paradigm. The design challenges have been considerable. First, high L/D vehicles have significantly higher ballistic coefficients, or β (β =M/CdA, where M is the mass, Cd the drag coefficient, and A the reference surface area). The longer flight times result in a much higher total integrated heat load (which scales directly with β). Secondly, the sharp features and control surfaces on such vehicles can undergo extremely high heating rates. If not properly designed, these features can ablate improperly, resulting in undesirable pitching moments (again, our solution to the above is a combination of reduced flight time, improved TPS and initial usage of a decelerator).

Thus, there is great attraction in the foregoing mission concept for the flight experiment component. This would allow for inexpensive flight testing of critical flight elements, and allow for the testing of the modified aerodynamic data base in the actual Mars flight environment. This tested modification of an existing data-base would be useful for whatever lifting vehicle configurations would follow. Finally, the choice of SWERVE-derived vehicle deserves some comment. First, and most obvious, is that it is extremely economical to modify an existing vehicle and data-base. A second and more subtle reason, however, is that due to the wings, the stall speed is reduced to below Mach 2 (proposed roll-modulated bi-cones, or even a mature system such as AMaRV, the Advanced Maneuverable Reentry Vehicle, exhibit stall at around twice the velocity, Reference 22). This allows for flight verification at lower Mach numbers, but also found to be useful in science missions where controlled flight at lower velocity is desirable.

Summary

A preliminary design of an integrated AEOLUS vehicle capable of precisely delivering a variety of scientific instruments to the surface of Mars has been presented. The complete package is approximately 4.5 meters long, 1.2 meters in diameter at the base, and weighs 778 kilograms. Not only does the AEOLUS concept design address a basic scientific need, but its development process also corresponds to a general philosophy emphasized within NASA to use innovative approaches to complete interplanetary exploration. The current status of the enabling technologies required for the AEOLUS system indicates that this could be accomplished in a relatively low-risk manner.

Some of the key areas presented in the proposed mission are:

- The development of a science rationale and investigation approach regarding specifically targeted sites of interest to the exobiology /exopaleontology community; initially by use of existing penetrator technology with the ability to evolve to more complex roving instruments.
- The demonstration of precision navigation resulting in the delivery of scientific instruments to within a 500 m radius or less.
- The use of a high L/D vehicle for planetary exploration (thus breaking a long standing paradigm).
- The development of a technology mission that is a stepping stone to future robotic and manned Mars missions.
- The attempt to solve the problem of high transportation costs which continue to plague planetary missions.
- The proposal of a scientifically provocative mission of modest cost, combining the above features, by making use of an array of existing technology.

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Biographical Data for Marcus S. Murbach

MarcMurbach has engineering degrees from Harvey Mudd College, San Jose State University and is currently pursuing a Ph.D. at Stanford University in the Aeronautics/Astronautics Department. Marc has worked for a number of years at the NASA Ames Research Center in California. Initially, he worked in the Thermo and Gas-Dynamics Division and most recently in the Space Projects Division. Work areas have included numerous advanced system studies and mission analysis, particularly related to problems of atmospheric entry. In the early 90's, he was the Ames representative during the Space Exploration Initiative and continues to have strong interests in both the robotic and manned exploration of Mars. He is the originator of 'Aeolus' system concept, and works on developing other atmospheric entry tools to expand space exploration. More recently, he is the principal investigator in the development of unique hypersonic experiments on sounding rockets. Marc enjoys providing public lectures on space history and exploration.

Biographical Data for David L. Keese

David Keese is currently the manager of the Rocket Systems Technology Department in the Aerospace Systems Development Center at Sandia National Laboratories. During his employment at Sandia, Mr. Keese has been involved in virtually all aspects of advanced aerospace vehicle analysis, design, development and flight testing. His experiences have included aerodynamic modeling of rocket and reentry systems, trajectory simulations to model the flight dynamics of these vehicles, and extensive interactions with test range personnel in connection with flight safety analyses. David's educational background includes graduate and undergraduate degrees from Texas A&M University in Aerospace Engineering. He also recently completed the Executive MBA program offered by the Anderson Graduate School at the University of New Mexico. Mr. Keese's interest in space activities led him to encourage the formation of an informal team of Sandians known as the Sandia Space Council. This group meets regularly to discuss common interests in space programs and to share information concerning new events and opportunities in this area. David also directed Sandia's early interactions with the Southwest Regional Spaceport to provide Sandia analysis expertise to support the initial feasibility study for a spaceport at the White Sands Missile Range.

Biographical Data for Jack Farmer

Jack Farmer received degrees in Geology from California State University, Chico (B.A.), the University of Kansas (M.S.) and U.C. Davis (Ph.D.) and has since held positions as Senior Museum Scientist and Lecturer (U. C. Davis), Senior Petroleum Geologist (Exxon, Western Division Production, Los Angeles), and Visiting and Adjunct Professorships in the Dept. of Earth and Space Sciences at UCLA. Jack joined NASA-Ames Research Center in 1991 as a National Research Council Senior Fellow. In 1994 he became a Civil Servant and Research Scientist in the Exobiology Branch. His interests include early biosphere evolution, the microbiology and biosedimentology of thermal springs and other extreme environments, and strategies for exploring Mars for a past or present biosphere. He is Associate Editor for "Origin of Life and Evolution of the Biosphere" and is active in the Geological Society of America, the American Geophysical Union, the Paleontological Society, and the American Indian Science and Engineering Society. He holds appointments with NASA's Solar System Exploration Sub-Committee, JPL's Microrobotics Science Advisory Group, the Solar System Exploration Roadmap Development Team, the Mars 2001-2005 Science Working Group, and the Mars 2001 Science Definition Team.